

Advanced Manufacturing Series
NIST AMS 100-69

**A Fully Registered In-Situ and Ex-Situ
Dataset for Metal Powder Bed Fusion
Additive Manufacturing**
*Data Processing, Feature Extraction, Registration, and
Uncertainties*

Zhuo Yang
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Zhuo Yang

Yan Lu

Ho Yeung

Brandon Lane

*1National Institute of Standards and Technology
Gaithersburg, MD 20899*

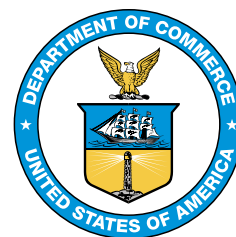
Nicole Van Handel

Georgia Institute of Technology

Atlanta, GA 30332

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Author ORCID iDs

Zhuo Yang: 0000-0001-7112-6266

Nicole Van Handel: 0000-0000-0000-0000

Yan Lu: 0000-0003-2927-5860

Ho Yeung: 0000-0002-6073-5709

Brandon Lane: 0000-0001-5153-1678

Abstract

This document details the data registration process for the previously published datasets from Additive Manufacturing Metrology Testbed (AMMT) parts, "Overhang Part X4," generated at the National Institute of Standards and Technology (NIST). The two datasets—one for process monitoring and the other for XCT inspection—covering four overhang parts, along with their descriptions were published in 2020. The published data have been well-received by the community, advancing the understanding of laser powder bed fusion additive manufacturing (AM). In the last four years, the NIST team encountered numerous questions regarding the published datasets, as the raw data were not easily interpretable for mining process-structure relationships. To support a wider range of research efforts across multiple disciplines, the NIST team conducted additional data analysis, resulting in a fully registered and well-documented dataset for publication. This document provides a detailed overview of the data processing pipeline and the multi-modal data registration techniques employed, including preprocessing, feature extraction, and data alignment. The final registered dataset consists solely of numerical values, fully aligned with the machine coordinate system. Key features of the registered data include process parameters, laser power, in-situ melt pool characteristics, in-situ layerwise optical intensity, and ex-situ XCT voxel values. Additionally, this document provides uncertainty analysis for each feature to help users better select data for their applications and evaluate their results. It can also serve as a framework for processing similar datasets collected on the same testbed in future research.

Keywords

Additive manufacturing; data management; data registration; feature extraction; uncertainty quantification.

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1. Introduction

This document details a fully registered dataset published at data.nist.gov. The full name of the dataset is the same to this paper, which can be downloaded from <https://data.nist.gov/od/id/mds2-3761>. This dataset integrates registered features extracted from multi-modality in-situ and ex-situ measurements of a laser powder bed fusion (LPBF) additive manufacturing (AM) process into single comma-separated values (CSV) files, organized in layer and per part. The original raw data were published in 2020, with all files downloadable from data.nist.gov, and an open access of data descriptions [1, 2]. Detailed information and links to the data are listed in Section 2. Users who utilize this dataset should also credit the original publications:

- Lane, Brandon, and Ho Yeung. "Process monitoring dataset from the additive manufacturing metrology testbed (AMMT): Overhang part x4." *Journal of Research of the National Institute of Standards and Technology* 125 (2020): 1-18.
- Praniewicz, Max, Brandon Lane, Felix Kim, and Christopher Saldana. "X-ray Computed Tomography Data of Additive Manufacturing Metrology Testbed (AMMT) Parts: Overhang Part X4." *Journal of Research of the National Institute of Standards and Technology* 125 (2020): 1-9.

Since the original datasets were published, they have been downloaded over 600 times as of time of this publication. Early interactions with users involved multiple rounds of providing meta-information and clarifying measurement data. Analyzing original multi-modality datasets require users to spend substantial effort on understanding the process, familiarizing oneself with the machine settings, addressing multi-modalities, and performing the time-consuming task of downloading and unzipping large files. The goal of this work is to offer a pre-processed and fully registered dataset that enables researchers to focus on solving domain-specific problems. The problems could be developing process-structure relationships, detecting process faults, and predicting part defects—rather than spending excessive time interpreting and formatting the raw data from the Additive Manufacturing Metrology Testbed (AMMT) at the National Institute of Standards and Technology (NIST) [3]. While the original raw data, including various types of images, offer valuable opportunities for research—such as deep learning-based part defect prediction—the registered dataset greatly enhances the user’s experience by providing clear, structured tabular data that is more accessible for most research activities.

This registered dataset contains only numerical values in CSV files, with features defined and extracted from excel sheets, coaxial images, layerwise images, and X-ray Computed Tomography (XCT) scans. Users can directly utilize the data for modeling, analysis, or machine learning. The following sections are organized as follows: Section 2 briefly introduces key information about the original experiment and raw data; Section 3 provides a full description of the rows and columns in the CSV file; Sections 4 and 5 describe the data processing and registration methods used to generate the data, respectively; Section 6 discusses the

uncertainties in the entire data; and the final section offers a summary with additional information.

2. Original Experiment and Raw Data

The original experiment was conducted on the NIST AMMT. This 3D build experiment was designed to produce four nominally identical parts within a single build while maintaining consistency in material, scan strategy, and laser processing parameters. The primary objective was to test the process consistency of multiple parts in one build. The study investigated the both in-situ and ex-situ measurement. For more details about the experiment, including part geometry, scan strategy, and sensor specifications, readers are encouraged to refer to Lane and Yeung, 2020, and Praniewicz et al, 2020. Original datasets were published and can be downloaded by the following links:

- Link of original in-situ dataset: <https://data.nist.gov/od/id/mds2-2233>
- Link of original ex-situ dataset: <https://data.nist.gov/od/id/mds2-2291>

Fig.1 illustrates the part geometry and Fig.2 shows the layout on the build plate. Each part measures $5\text{mm} \times 5\text{mm} \times 9\text{mm}$ and consists of 250 layers. A 90° rotation is applied between each successive layer. The scan sequence follows a sequential order, progressing from Part 1 to Part 4, with each part being fully completed before advancing to the next layer.

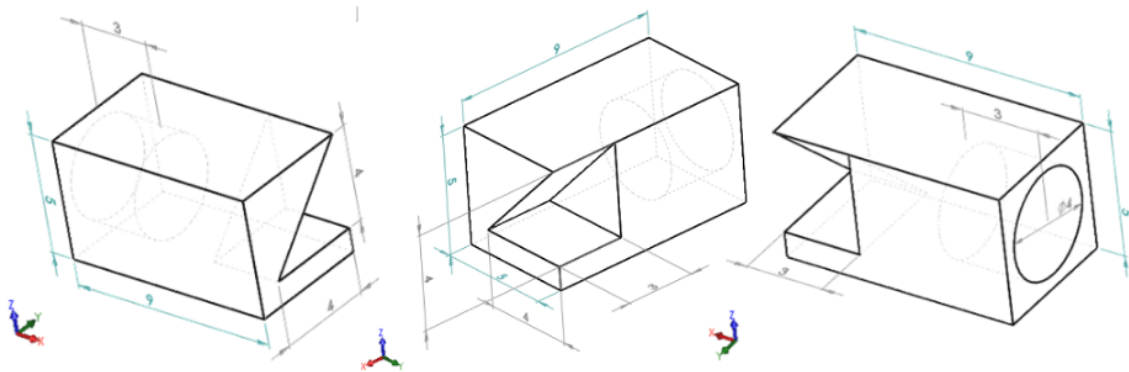


Fig. 1. Part geometry in three views [1].

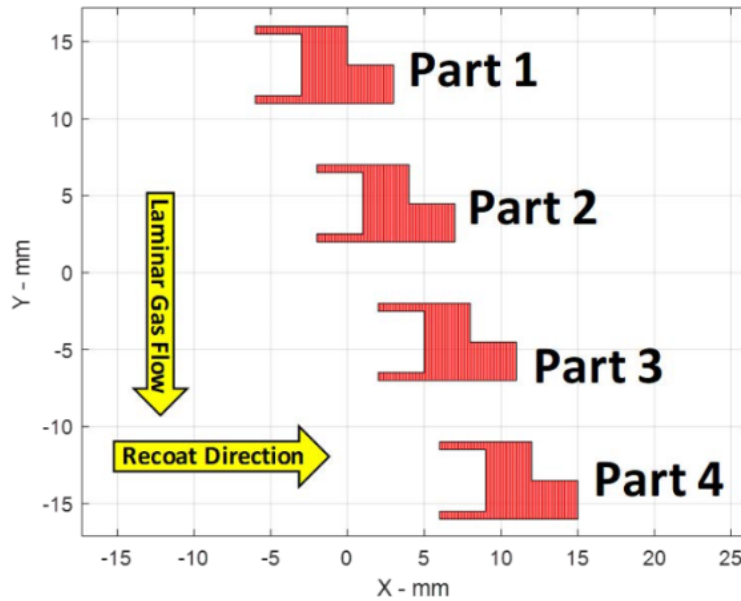


Fig. 2. Part numbering and layout for Layer 125 of 250 [1].

The original datasets comprise five components:

1. Digital commands from the scan file, namely the XYPT command, Laser position in X axis nad Y axis, Laser Power, and Camera Trigger.
2. Data Acquisition (DAQ) data from the Galvo X/Y-mirror encoder,
3. Melt Pool Monitoring (MPM) images captured by a coaxial high-speed camera.
4. Layerwise Images (LWI) taken with three illumination LED (A, B, and C) positions.
5. Micro-XCT scan data

Detailed information about the sensors, their configurations, and raw data processing methods can be found in the data description provided in the original dataset release.

3. Data Description

3.1. Data Structure

The data structure is shown in Fig.3 that the registered X4 dataset has four folders. Each folder refers to one part with 250 CSV files, where each CSV file has all features extracted from the 5 components of one layer.

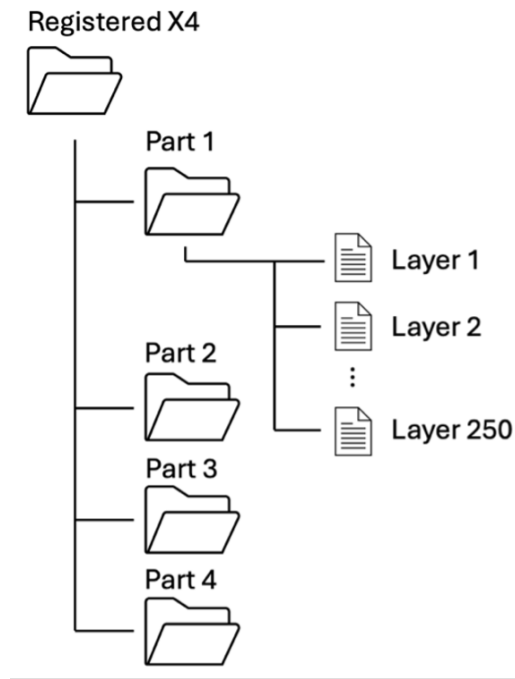


Fig. 3. Data structure of registered X4 dataset.

3.2. Registered Features

Each CSV file is formatted into 40 columns and multiple rows. Each row represents a measured point with all associated features, as listed in Table 1, 2, and 3.

4. Data Processing Method

This section introduces the detailed methods for raw data processing. These steps are fundamental preparation for data registration. Such steps include calculating the scan speed from laser position, image process for feature extraction from raw MPM images, shape correction of raw LWI, and interpolating of original XCT data.

4.1. Scan Speed

The scan speed is calculated based on the laser position and time interval. Equation 1 is used to determine the raw speed from both command and real position.

$$v = \frac{\sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}}{t_i - t_{i-1}} \quad (1)$$

Table 1. Features of the dataset.

	Name	Unit	Definition
1	Part number	N/A	Same as scanning order
2	Build time	μs	Synchronized with machine time
3	Command laser position in X axis	mm	Extracted from XYPT command
4	Command laser position in Y axis	mm	Extracted from XYPT command
5	Command laser power	W	Extracted from XYPT command
6	Command scan speed	mm/s	Calculated based on command laser position
7	Real laser position in X	mm	Extracted from DAQ
8	Real laser position in Y	mm	Extracted from DAQ
9	Real laser power	W	Extracted from DAQ
10	Real scan speed	mm/s	Calculated based on real laser position
11	Melt pool length	mm	Major axis of fitted ellipse, threshold 80
12	Melt pool width	mm	Minor axis of fitted ellipse, threshold 80
13	Melt pool area	mm^2	Melt pool area, threshold 80
14	Melt pool length	mm	Major axis of fitted ellipse, threshold 100
15	Melt pool width	mm	Minor axis of fitted ellipse, threshold 100
16	Melt pool area	mm^2	Melt pool area, threshold 100
17	Melt pool length	mm	Major axis of fitted ellipse, threshold 120
18	Melt pool width	mm	Minor axis of fitted ellipse, threshold 120
19	Melt pool area	mm^2	Melt pool area, threshold 120

Here, x_i and y_i denote the laser positions at time t_i , while x_{i-1} and y_{i-1} denote the laser positions at time t_{i-1} . It is important to note that i and $i - 1$ correspond to the i_{th} and $(i - 1)_{th}$ rows in the original XYPT command, representing two successive laser positions. Consequently, the time difference $t_i - t_{i-1}$ is consistently $10 \mu s$. In this registered dataset, two successive rows indicate two consecutive measured positions, which has a minimum $10 \mu s$ difference. Two rows may have larger time interval, over $10 \mu s$, while the laser is turned off and no data was captured. As a result, the two points may not necessarily lie on the same scan track.

The command scan speed (Row 6 in Table 1) is derived directly from Equation 1. To calculate the real scan speed (Row 10 in Table 1), a 1D mean filter with a window size of 3 is applied to account for the measured noise.

4.2. Melt Pool Feature Extraction

Melt pool dimensions were extracted through the measurement of binarized melt pool images. Extracted features of the melt pool can be affected by the measurement. For example, calibrated temperatures do not directly correspond to the true conditions of melt pools but the optical signal from the camera. Phenomena such as plume and spatter can

Table 2. Features of the dataset (continued).

	Name	Unit	Definition
20	LWI pixel value	N/A	Powder view after spreading, with LED A, original value
21	LWI pixel value	N/A	After spreading powder view with LED A, 3×3 mean filter applied
22	LWI pixel value	N/A	After spreading powder view with LED A, 5×5 mean filter applied
23	LWI pixel value	N/A	After spreading powder view with LED B, original value
24	LWI pixel value	N/A	After spreading powder view with LED B, 3×3 mean filter applied
25	LWI pixel value	N/A	After spreading powder view with LED B, 5×5 mean filter applied
26	LWI pixel value	N/A	After spreading powder view with LED C, original value
27	LWI pixel value	N/A	After spreading powder view with LED C, 3×3 mean filter applied
28	LWI pixel value	N/A	After spreading powder view with LED C, 5×5 mean filter applied
29	LWI pixel value	N/A	Exposure view after scan, with LED A, original value
30	LWI pixel value	N/A	Exposure view after scan, with LED A, 3×3 mean filter applied
31	LWI pixel value	N/A	Exposure view after scan, with LED A, 5×5 mean filter applied
32	LWI pixel value	N/A	Exposure view after scan, with LED B, original value
33	LWI pixel value	N/A	Exposure view after scan, with LED B, 3×3 mean filter applied
34	LWI pixel value	N/A	Exposure view after scan, with LED B, 5×5 mean filter applied

cause the perceived melt pool to appear much larger than its true shape. Without additional channels of data, there are no established methods to verify the true dimensions of the in-process melt pool by pure optical images. Furthermore, much of each melt pool image registers as above the melting temperature of the material. Thermal calibration can be referred to the original experiment [4, 5].

Table 3. Features of the dataset (continued).

	Name	Unit	Definition
32	LWI pixel value	N/A	Exposure view after scan, with LED B, original value
33	LWI pixel value	N/A	Exposure view after scan, with LED B, 3×3 mean filter applied
34	LWI pixel value	N/A	Exposure view after scan, with LED B, 5×5 mean filter applied
35	LWI pixel value	N/A	Exposure view after scan, with LED C, original value
36	LWI pixel value	N/A	Exposure view after scan, with LED C, 3×3 mean filter applied
37	LWI pixel value	N/A	Exposure view after scan, with LED C, 5×5 mean filter applied
38	XCT voxel value	N/A	Original value
39	XCT voxel value	N/A	$3 \times 3 \times 3$ mean filter applied
40	XCT voxel value	N/A	$5 \times 5 \times 5$ mean filter applied

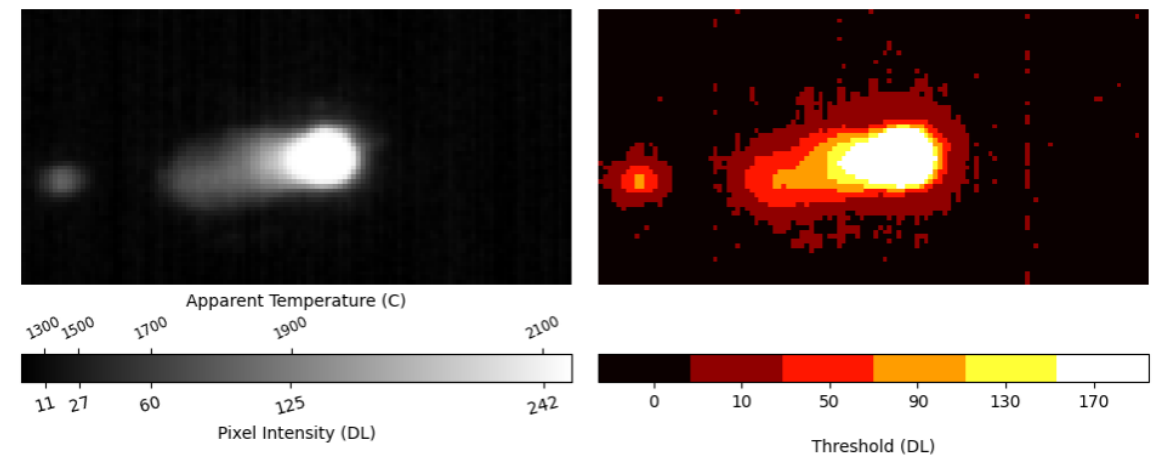


Fig. 4. (Left) the raw grayscale melt pool is shown with a color bar describing the pixel intensity values. Intensity values are mapped to measured apparent temperature via the Sakuma-Hattori curve described in the data description. (Right) the same image is shown mapped based on different intensity thresholds.

Some methods propose using histogram separation to measure the melt pool [6]. However, this will result in different apparent temperature regions being measured, depending on the image. Fig. 4 shows the nonlinear mapping between temperature and image intensity, along with the resulting image at different value cutoffs.

As the cutoff value increases, the resulting melt pool becomes smaller. The optimal range for value separation was found to be between 80 and 120 based on previous experience. Therefore, in this dataset, the measured melt pool is reported for binary images resulting from thresholds of 80, 100, and 120. These cutoffs correspond to calibrated temperatures of approximately 1800°C, 1850°C, and 1900°C, respectively.

Melt pool dimensions are derived in the following steps:

1. The raw image is thresholded using a specified cutoff value.
2. The largest connected component of the binary image is isolated.
3. Melt pool edge coordinates are extracted using the Sobel edge detection method.
4. Ellipse fitting is performed on the edge points using the least squares method.

The steps are shown in Fig. 5, with a comparison of measurements obtained using different threshold values. Steps 1 and 2 denoise and isolate the primary melt pool. Steps 3 and 4 perform a best-fit to an ellipse [7]. The ellipse axes are used to measure the melt pool length and width, where the length is assumed to be the major axis and the width the minor axis. Finally, the area is reported by summing the pixels in the largest connected component image. Given that the pixel size is a constant $8\mu m \times 8\mu m$, all measurements can be converted from pixels to microns. A discussion on the measurement uncertainty of this methodology is provided in Section 6.

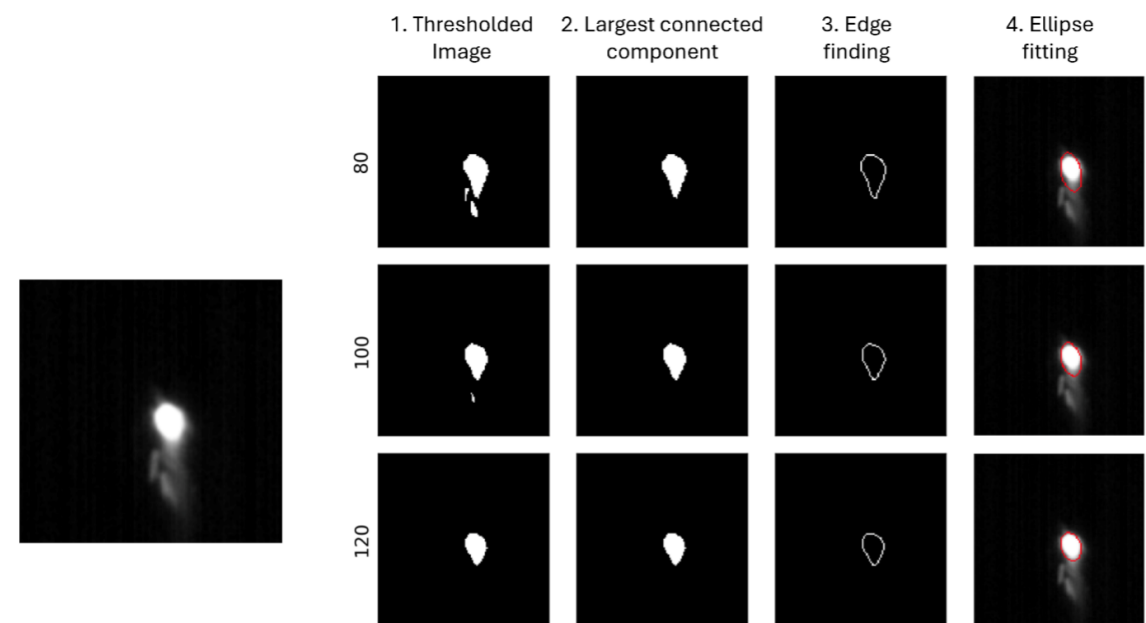


Fig. 5. (Left) Raw melt pool image, (right) the feature extraction process is shown for 3 difference threshold levels.

4.3. LWI Processing

To report image values, LWI were corrected for perspective shift and aligned [8, 9]. The images were transformed using a 3D projection matrix (M) that maps the observed corners to the aligned corners of a 180×100 pixel rectangle, as shown in Equation 2.

$$\begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = M \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} \quad (2)$$

As shown in Fig.6, the process begins with cropping each part from the raw LWI. Four corner points, p1 to p4, were selected manually based on observation on Layer 17, whose cross-section is $5\text{mm} \times 9\text{mm}$ rectangular as designed. These selected points are paired with the designed part geometry. As depicted in Fig.6(c), the four points in XYPT provided the locations and shape of the cross-section. During the build, the camera position remains fixed, and the height of the build plate remains constant. To simplify the process, the projection matrix is calculated based on one layer and then applied to all subsequent layers. Layer 17 (L0017C in the original LWI files, where “C” represents to LED C) is selected to pick the four points for all parts since it has the clearest edges.

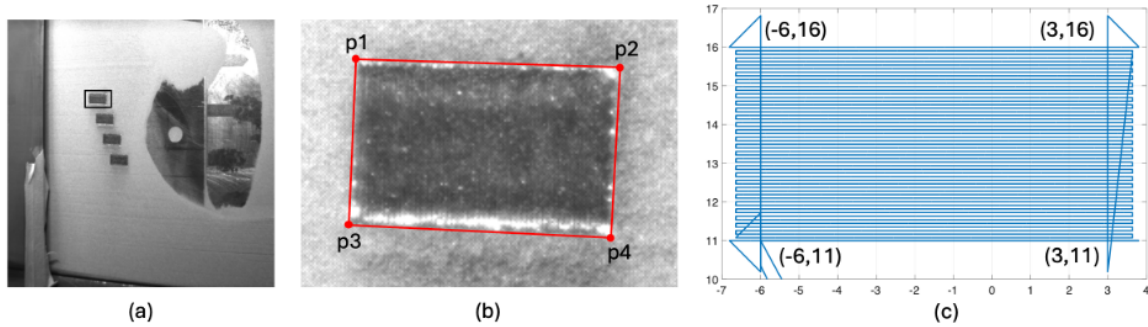


Fig. 6. Points selections. (a) part is cropped from the raw LWI image. (b) 4 corner points were manually selected. (c) dimensions were captured from the XYPT command.

The shape correction process is shown in Fig.7. Following these steps, the image is cropped to the bounding area, resulting in the image origin being aligned to the commanded origin. This transformation was applied to all LWI per part. Intensity values were reported for the raw image as well as the mean filtered images. The corrected LWI, after removal of non-scanned area, is 180×100 pixel.

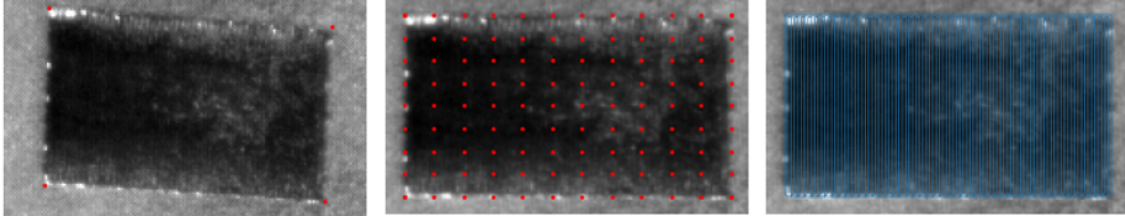


Fig. 7. The perspective transformation of a given image and corresponding coordinate alignment. The original cropped image (left). The aligned image with dotted points (middle). The XYPT commands are projected onto the corrected image (right).

4.4. XCT Processing

The voxel size of the original XCT is $11.95\mu m$, indicating that the pixel size in the original TIFF image is $11.95\mu m \times 11.95\mu m$. Additionally, the distance between two consecutive XCT slices is $11.95\mu m$. Given the $20\mu m$ layer thickness, it is challenging to align most XCT slices with the build layers. To resolve this issue, the first step is to remove the original XCT slices without volume incorporating with real measurement. Since the 4 parts have different dimensions, this removal may result in the XCT measurement not consistent between each other. The second step is to resize the XCT data to a $10\mu m$ voxel size using the cubic interpolation method.

5. Data Registration Method

Data registration here refers to the integration of multiple datasets spatially and temporally. The fundamental reference is the XYPT command, to which all features are registered. Each row includes all features corresponding to that command position where the MPM is captured. This section introduces all techniques used to align the extracted features from the raw datasets. The XYPT command, as the original reference, does not require any alignment. The only operation performed is to discard rows in the XYPT without captured MPM. Since the coaxial camera operates only when the laser is turned on at a $10kHz$ rate, the registered data cannot fully reflect the laser scan path. It cannot cover the path that laser is off since no physical part or measurement are performed. Users can examine the original XYPT to explore the scan path in detail.

5.1. DAQ to XYPT

The DAQ reflects the real laser position and laser power, but it is captured by the encoder embedded in the Galvo and does not include the camera trigger. This prevents the direct alignment of the real laser position with the XYPT. The alignment is made based on the laser behavior [10]. Given the delay when switching the laser from $0W$ to a positive value, the laser requires a short warm-up period. In the DAQ, when the laser is not on, the average

power is negligible. When the laser turns on, the laser power in XYPT immediately changes to 100W or 195W. However, in reality, the actual laser power gradually increases from around 0 to near the designed value [11]. Observations from MPM show that the first image of each track has no melt pool. As a result, the first positive laser power in the DAQ at the beginning of the first track is registered to the first positive laser power in XYPT. This process is repeated for each layer. The alignment is assumed to be consistent throughout the layer, assuming no additional delays occur during scanning. All tracks are checked to ensure that registration would not cause obvious issues. Such issues include melt pool should not be presented while laser is off.

5.2. MPM to Laser Position

MPM images are directly registered to the command position using the camera trigger in the XYPT [12, 13]. Fig.8 shows the position of camera triggers on the scan path that cropped from actual data.

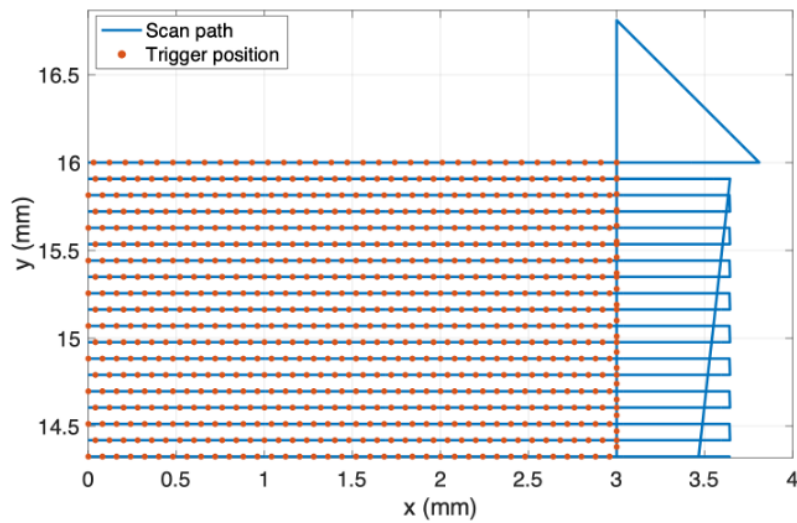


Fig. 8. Scan path and camera trigger position.

5.3. LWI Pixel to Laser Position

The LWI is registered to the part geometry based on the XYPT laser position. For each part, the outer area of the processed LWI is cropped. The top-left point of each part in XYPT is assigned to the top-left pixel. For each measured location (row in the file), the nearest pixel is selected for registration. Fig.9 shows the registered LWI in the machine coordinate system.

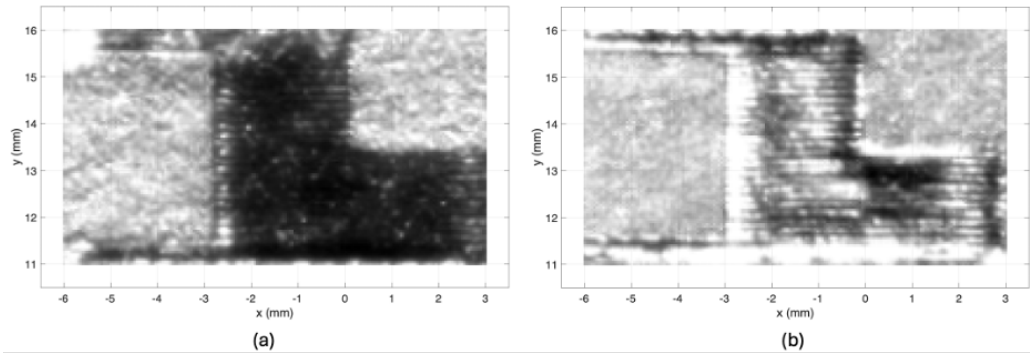


Fig. 9. Registered LWI images. (a) Layer 126, Part 1, LED C. (b) Layer 126, Part 1, LED 3.

Many factors can affect the registration accuracy, with errors potentially originating from perspective transformation and pixel selection. These errors are difficult to avoid. To mitigate this, different filters are applied to ensure the registration is generally accurate. Two mean filters, 3×3 and 5×5 , were added in addition to the originally processed LWI. Fig. 10 (LED B) and Fig. 11 (LED C) show the comparison between the filtered and original LWI. After filtering, the registration can tolerate a misalignment of up to 4 pixels.

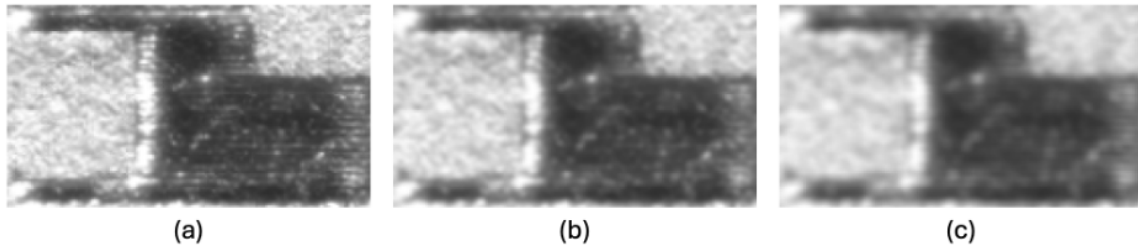


Fig. 10. Registered LWI image, Layer 165, LED B, (a) without mean filter, (b) 3×3 mean filter, and (c) 5×5 mean filter.

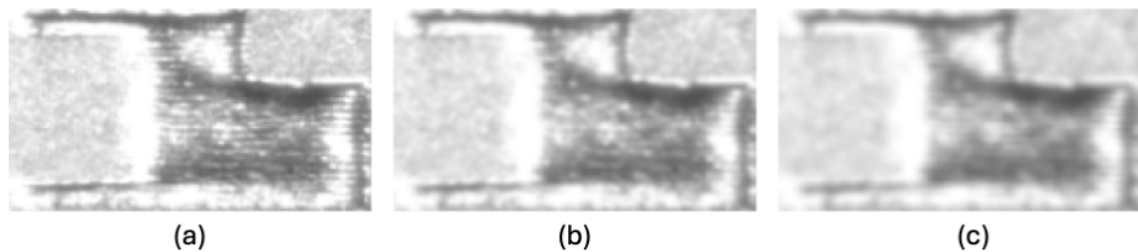


Fig. 11. Registered LWI image, Layer 165, LED C. (a) without mean filter. (b) 3×3 mean filter. (c) 5×5 mean filter.

5.4. XCT Pixel to Laser Position

The 3D volume of rescaled XCT data is registered to the build layers using three pairs of reference surfaces along the x, y, and z axis. Since the surfaces of the final build are not perfectly flat, it is challenging to align the XCT surface to the build layer. The same method has been approved reliable for surface roughness measurement [14]. To achieve the best alignment, the registration process uses the best-fitted result of two reference surfaces in each direction. Fig.12 shows the alignment in each a xis. Two 3D mean filters, $3 \times 3 \times 3$ and $5 \times 5 \times 5$, are applied to accommodate alignment errors.

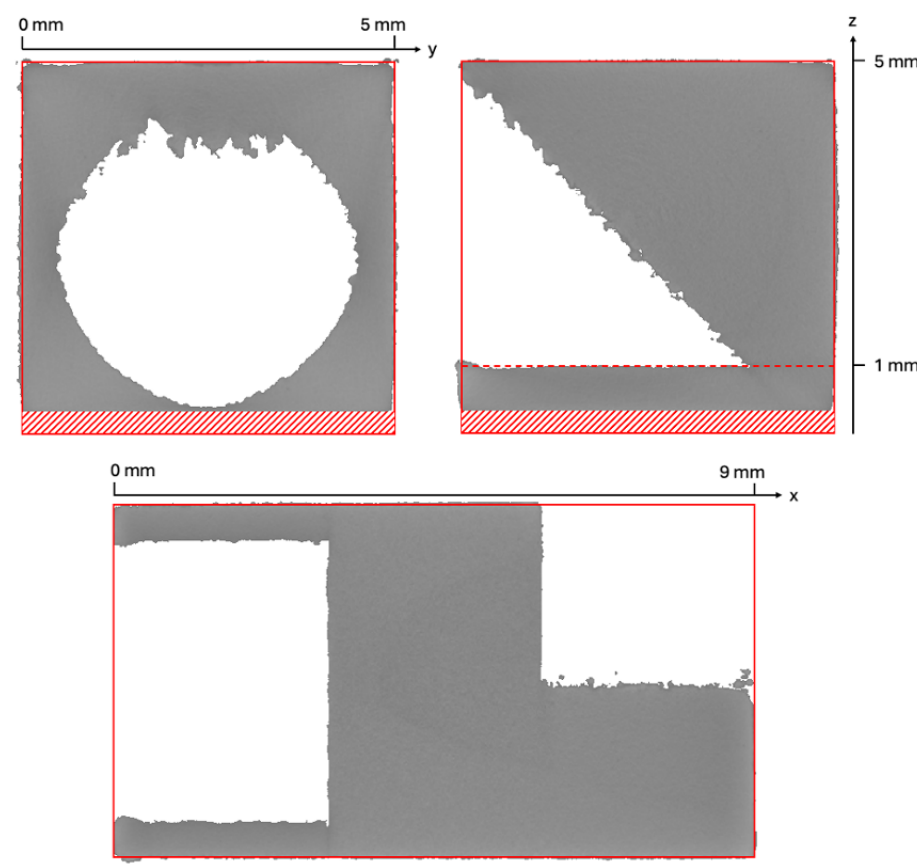


Fig. 12. XCT alignment in X, Y, Z axis. Both sides of each axis are selected to be the reference surfaces. The alignment is the best fitted for both sides. Since the bottom (shaded area) is removed partially from the build plate, z direction uses an alternative plane (dashed line) as reference.

6. Uncertainty Analysis

Each reported measurement contains all uncertainties contributed by our processing methods to inform users of sources of error that may affect future model construction. Uncer-

tainties from data collection methods may be referenced from prior publications, while relevant sources are noted as they pertain to the methods in this work.

6.1. Melt Pool Features

Melt pool feature extraction propagates uncertainties from the thermal cameras used as well as the measurement processes described. This work defines the measurands of the melt pool dimensions with respect to given apparent temperatures. A related discussion involving uncertainty on AMMT melt pool cameras can be found in related works and the original data description [5, 15, 16], wherein track lengths were measured. Users may use this work as a reference to translate apparent temperature into true temperature.

Camera artifacts of signal noise, signal digitization, and blur all impact the certainty of melt pool measurements. Representing a melt pool in pixels wherein light exists at quantized levels results in a length uncertainty of 1 pixel (spatial pixelation) and an intensity uncertainty of 1 discrete/digital level (DL). Given that uncertainty is typically assumed to be uniform, both values may be assumed uniformly distributed. Motion blur describes the apparent lengthening of an image target in the direction of motion; the effect is a result of the target moving over the span of integration time. A conservative upper bound can be found for length measurements using an estimation model, for example, $2vt$ where v is travel speed and t is integration time. In this case, image integration time is $20\mu\text{s}$ and that the travel speed differed between contour and infill as 800mm/s and 900mm/s respectively. Users may elect to consider the effect of motion blur on the width as negligible. An approximation for optical blur, resulting from a suboptimal lens focus, was developed previously via the knife-edge technique and found to be $2.9\mu\text{m}$, distributed normally, in a prior work.

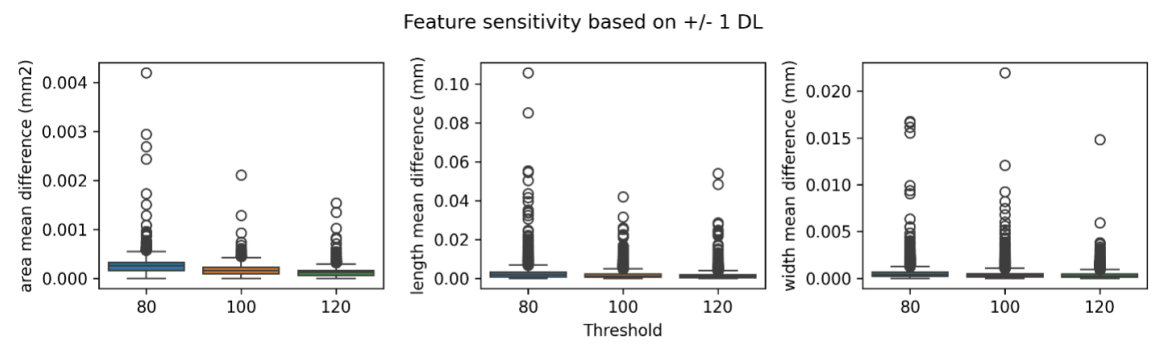


Fig. 13. Boxplots of melt pool area, length, and width sensitivity from varying the threshold by ± 1 DL, measured on Layer 1.

Measured melt pool features are based on the chosen threshold. However, the intensity at each pixel should have an assumed uncertainty of ± 1 DL from signal digitization and camera noise, with uniform probability. That can define an upper and lower bound for the

melt pool dimensions for a given temperature reading by varying the threshold about a given value. In the case of the provided values, average centered difference between the measured values for a melt pool were used to determine measurement stability. Values are reported for the first layer, where observations are assumed to be generally independent. From Fig.13 and Table 4, variability from intensity variation contributes less to uncertainty than camera artifacts, which has a minimum uncertainty due to the pixel iFOV of $8\mu m$.

Table 4. Summary statistics of feature sensitivity based on the centered difference around a threshold of $\pm 1DL$

Feature	Area (mm^2)			Length (mm)			Width (mm)		
	80	100	120	80	100	120	80	100	120
Mean	0.0002	0.0002	0.0001	0.0023	0.0016	0.0013	0.0005	0.0004	0.0003
St. Dev	0.0001	0.0001	0.0001	0.0023	0.0015	0.0014	0.0005	0.0004	0.0003
Max	0.0042	0.0021	0.0015	0.1056	0.0420	0.0540	0.0168	0.0219	0.0148
Median	0.0003	0.0002	0.0001	0.0018	0.0011	0.0009	0.0004	0.0003	0.0003

6.2. DAQ Alignment

Encoder-recorded values for X, Y, and Power were aligned to the commanded values through using the power channel. Discussion on the error in encoder values is provided in the original data description, where compensation methods are provided. As in prior works, timing accuracy is assumed to be relatively accurate [1]. Values in this dataset that are reported are those tied to instances with associated melt pool images. Uncertainty due to registration may be assumed to be negligible relative to the measurement uncertainty of the encoder itself; alignment based on the first spike in power results in the lowest difference between commanded and aligned values for all channels, as shown in Fig.14 and Fig.15.

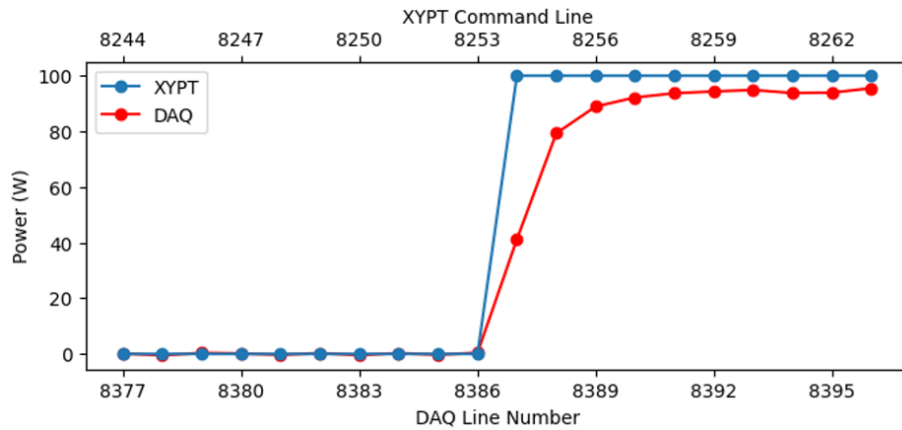


Fig. 14. Power values for XYPT and DAQ aligned by the first increase in power. Axes report line numbers for commanded and as-recorded data. Data points are separated by $10\mu s$ increments.

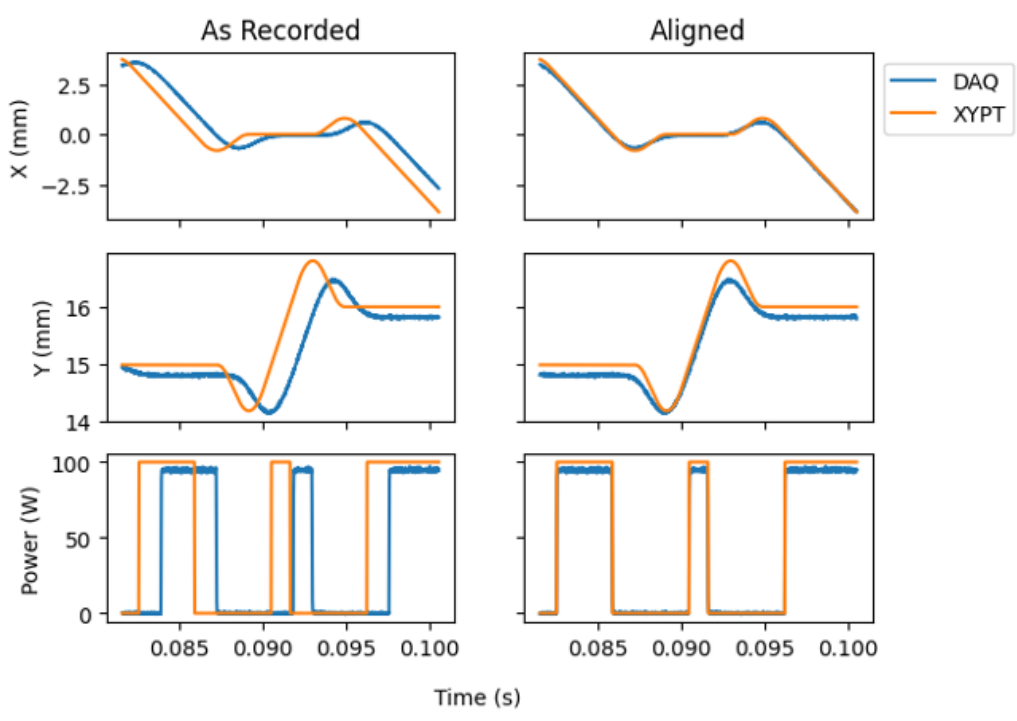


Fig. 15. X, Y, and Power channels for layer 200 (left) DAQ channels are plotted as recorded, (right) DAQ and XYPT channels are aligned to match based on the first spike in power [1]

6.3. LWI Alignment

For LWI, uncertainty mostly derives from the camera itself and from our alignment choice. In this dataset, intensity values are reported for the build at each lighting condition, before and after lasing. The measurand of image intensity has uncertainty resulting from the camera properties and the registration process. Lengthwise measurements are subject to artifacts such as spatial pixelation, while intensity values are affected by signal digitization, quantization, and noise. Camera characteristics are reported in the original data description.

LWI were subjected to geometric transformations to correct for the angle of the camera relative to the build plate, as described in Section 5.3. Measurement error may result from the selection of the corner selection. Due to the reflection on the captured image and geometric error of the build, there is potential error while selecting the points. An example of error resulting from poorly chosen corners is given in Fig.16 where the optimal selection is in red. The resulting coordinates from a transformation with perturbed corners is shown. The transformation resizes the build area so that the part measures 180×100 pixels, such that each pixel represents a $5\mu m \times 5\mu m$ area.

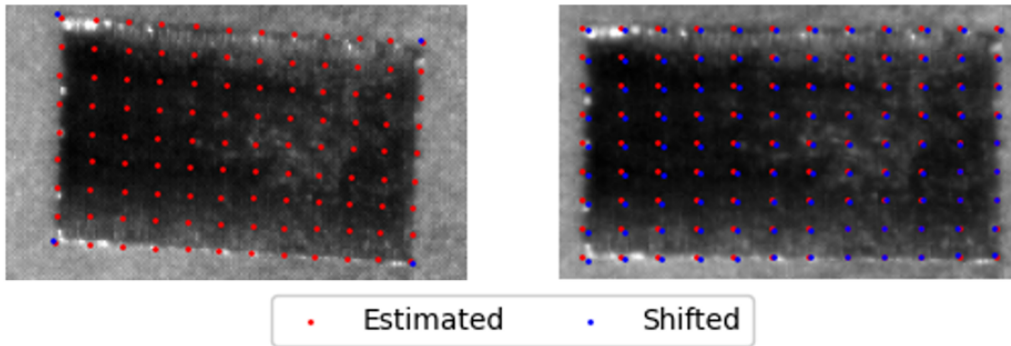


Fig. 16. The effect of a slight perturbation in the choice of corners is compared. Two choices of corners are given in red and blue, where red is a good estimation of the corner positions while blue is the same set of points perturbed a small distance.

The error for any given point can be found by finding the distance between the “true” points and those projected by shifted points. Error distribution is related to the relative direction of perturbations but may be assumed to be normally distributed. Assuming a confidence in corner selection of 1 pixel, the error in the corresponding transformed image will not exceed 2 pixels.

Positional error translates to value uncertainty as the “true” value of the position of interest may exist anywhere within 2 pixels. The uncertainty at any given pixel is represented by the variability within this span. The distribution of uncertainty for all of the images taken in Layer 10 is given in Fig 17; the original images can be found in Fig.17 The uncertainty of pixel value decreases on average when values are taken from blurred images, as neighborhoods become more uniform upon blurring. The effect is more pronounced in the higher contrast images.

6.4. XCT Alignment

All XCT measurements are subject to uncertainties resulting from imaging conditions and reconstruction methods described in the original data description [2]. Inherent uncertainties from collection include those from the initial voxel size and bit depth of 16-bit. Additionally, steps for value correction should be considered. For this dataset, uncertainty has been further introduced by voxel downscaling and registration error. For each reported pixel, distance uncertainty is 1.2 voxel, which impacts both distance measurements and intensity measurements. Rescaled, each voxel represents $10\mu m$ uniformly, which puts a lower bound on the precision of any distance measurements.

Coordinate system registration is not trivial, as the commanded laser movement is programmed pointwise. However, each resulting track results in a 3D volume, which makes alignment from commands to the final volume challenging. The flat surfaces of the parts were chosen to perform alignment. The mean plane of each pair of sides was chosen

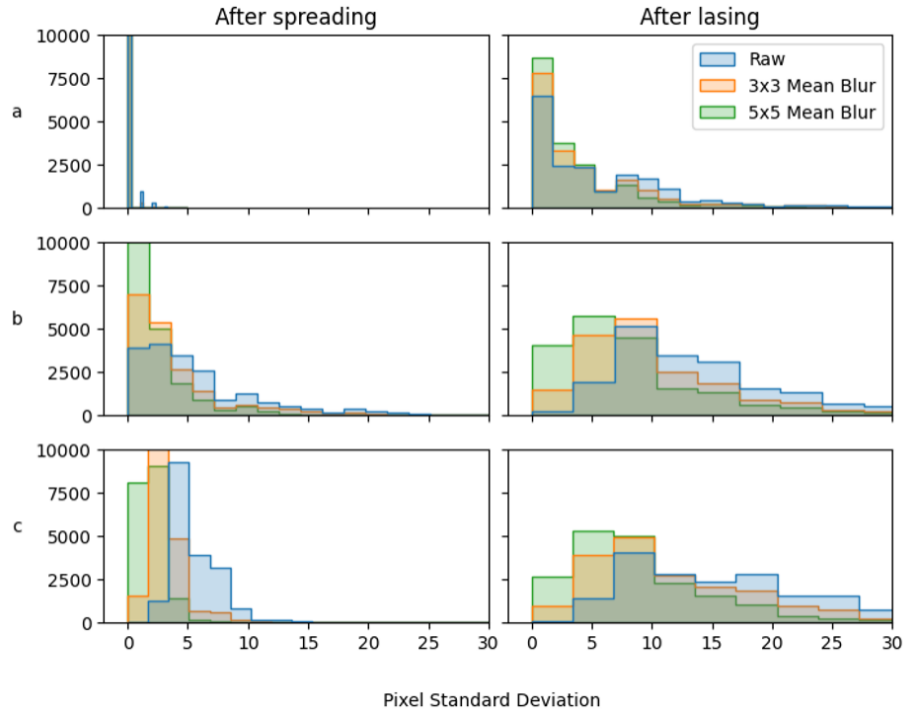


Fig. 17. Histograms for pixel standard deviation in a 5×5 neighborhood for all LWI taken during Layer 10. Symbol a, b, and c represent the LED flash conditions.

to align their respective axes, however, there is uncertainty in this selection, as surface roughness may bias the chosen plane. To determine a bound on the z value, true plane was estimated between the first and third quartile of all z values. The maximum uncertainty from datum alignment was found to be at most $\pm 10 \mu m$. Of the alignment surfaces, the top surface is the roughest; which may use this as an upper bound for all surfaces. Registration error may assume to be normally distributed. Table 5 lists the result.

Table 5. Top surface summary statistics. Points are sampled from the surface within $0.5mm$ of all sides to eliminate edge influences.

	Part 1	Part 2	Part 3	Part 4
Total Points	297427	251547	267433	254649
Min	4.567	4.549	4.540	4.542
Q1	4.601	4.593	4.572	4.567
Q2	4.608	4.605	4.577	4.571
Q3	4.615	4.614	4.582	4.578
Max	4.723	4.827	4.632	4.634
(Q3-Q1)/2	0.007	0.010	0.005	0.006

XYPT positions were matched to XCT values through interpolating across the z axis evenly for the nearest position match. However, part removal process from the build plate may have removed some layers from the process; this is best demonstrated by the X and Y dimensions exceeding the designed sizes but the part height being less than designed, as shown in Table 6.

Table 6. Part measurements based on the generated STL maximum and minimum values.

	Part 1	Part 2	Part 3	Part 4
X-extend (mm)	9.067	9.062	9.136	9.144
Y-extend (mm)	5.095	5.154	5.114	5.126
Z-extend (mm)	4.633	4.677	4.642	4.617

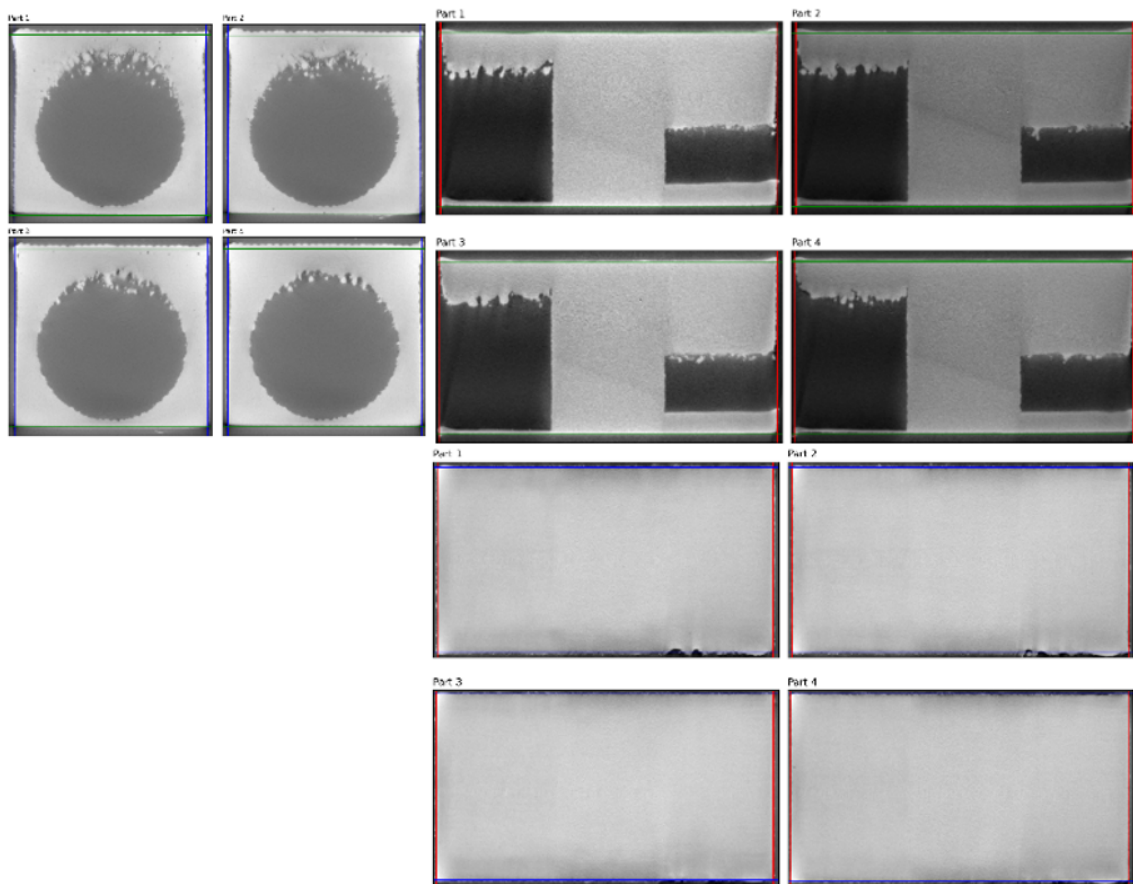


Fig. 18. Alignment datums are plotted onto the XCT images.

There is an additional component of uncertainty in the Z-direction (z-misalignment), which is from the possible misalignment due to missing layers. The z-misalignment would be

equivalent to $(z_0 - d)/250$ where z_0 is the designed dimension and d is the actual part dimension. In the ideal case, z_0 is 5mm , however, realistically, the z dimension would experience a similar expansion to the other dimensions. Next step is to consider the difference in misalignment based on the worst-case scenario, where the z -expansion is the same as the maximum observed expansion of the part. Based on $(5\alpha_{\max} - d)/250$, the maximum alignment error was $0.8\mu\text{m}$, which is negligible in comparison to the voxel size.

Distance uncertainty impacts reported intensity values as the voxel of interest lies within the neighborhood of uncertainty. The exact uncertainty value of a voxel can be found through measuring the variability in a 3D neighborhood. Similar to layerwise images, raw and mean filtered values are given for each location; using values from the filtered volume reduces the voxelwise uncertainty in the same way, shown in Fig.18 and Fig.19.

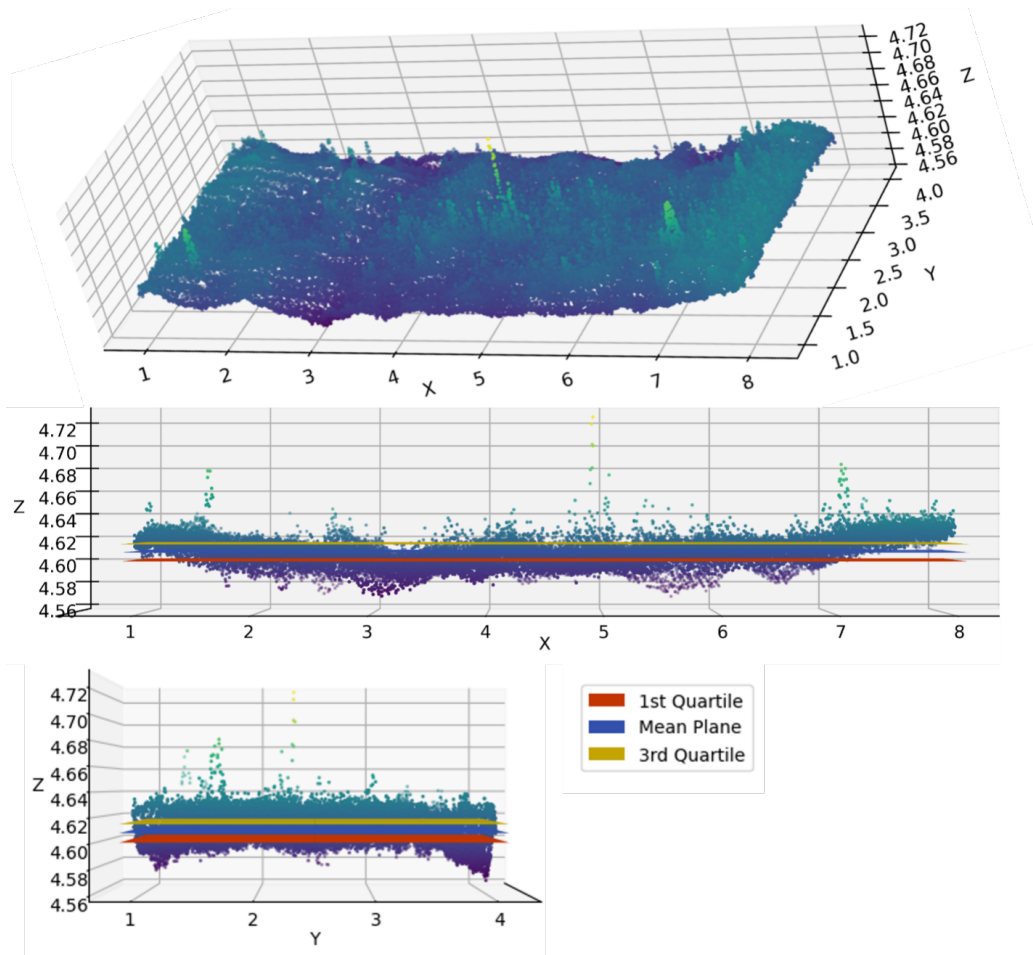


Fig. 19. The top surface topography of Part 1 XCT (down sampled). Planes are shown based on the quartiles of z values for the surface.

7. Summary

This description describes the works of generating a pre-processed, fully registered dataset in CSV format with numerical values extracted from various imaging modalities, allowing researchers to focus on tasks like process-structure analysis, fault detection, and defect prediction without extensive data preparation. The dataset improves accessibility for modeling, analysis, and machine learning, with clear documentation on its contents, processing methods, and associated uncertainties. More detailed information, such as raw images, can always be accessed via the original data files.

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